

The longest ring embedding in faulty hypercube *

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Abstract

In this paper, we show that the adjacency fault tolerance for property $2H$ of hypercube. We also show that the adjacency fault tolerance for Hamiltonian laceability of hypercube. Applying these results, we can construct the fault-free cycle with length $2^n - 2 \mid F_v \mid + 4$ in $Q_n - F_v$ where F_v is the faulty vertices set with at least two black vertices and two white vertices with $\mid F_v \mid \leq n - 1$.

1 Introduction

The hypercube network is the most popular interconnection network. It has many attractive properties, such as regularity, symmetry, recursively constructed. In this paper, we investigate the adjacent vertices fault tolerance for Hamiltonian properties of hypercube. We also consider the problem of embedding fault-free ring in hypercube when there are faulty vertices.

A bipartite graph $G = (V_b \cup V_w, E)$ is a graph such that $V_b \cap V_w = \emptyset$ and $\forall e = (u, v) \in E$, $u \in V_b$ and $v \in V_w$. Let $V_b(V_w)$ be the set of black(white) vertices. An n -dimensional hypercube Q_n is a bipartite graph with 2^n vertices. Each vertex is denoted as an n -bit binary string. Two vertices are adjacent if and only if their binary string representations differ exactly in one bit position. The hypercube Q_n can constructed recursively as $Q_n = Q_{n-1} \times K_2$. We can separate Q_n as two subgraphs Q_{n-1}^0 and Q_{n-1}^1 by choosing any one bit of binary string. We also denote $d(x, y)$ as the distance between vertices x and y .

Let $V(P)$ be the vertex set of the path P . Two paths P_1 and P_2 are *disjoint paths* if $V(P_1) \cap V(P_2) = \emptyset$. Two disjoint paths P_1 and P_2 of

$G = (V, E)$ are *spanning disjoint paths* if $V(P_1) \cup V(P_2) = V$. A *Hamiltonian cycle(path)* is a cycle(path) that visits every vertex exactly once. A graph G is *Hamiltonian* if G contains a Hamiltonian cycle. A graph G is *k edges Hamiltonian* if $G - F_e$ is Hamiltonian $\forall F_e \subset E$ and $\mid F_e \mid = k$. A bipartite graph $G = (V_b \cup V_w, E)$ is *Hamiltonian laceable* if there exists a Hamiltonian path between each pair of vertices u and v for $u \in V_b$ and $v \in V_w$. A Hamiltonian laceable graph $G = (V_b \cup V_w, E)$ is *hyper-Hamiltonian laceable* if $\forall v \in V_b(V_w)$, there exists a Hamiltonian path of $G - \{v\}$ between each pair of vertices of $V_w(V_b)$. A graph G is *k edges Hamiltonian laceable* if $G - F_e$ is Hamiltonian laceable $\forall F_e \subset E$ and $\mid F_e \mid = k$. A graph G is *k edges hyper-Hamiltonian laceable* if $G - F_e$ is hyper-Hamiltonian laceable $\forall F_e \subset E$ and $\mid F_e \mid = k$. In [6], Tsai et al. proved that Q_n is $(n - 2)$ edges Hamiltonian laceable and $(n - 3)$ edges hyper-Hamiltonian laceable.

There is no literature about general vertex fault tolerant Hamiltonian properties of hypercube for the number of faulty vertices more than one. Some literature concerns embedding a fault-free cycle for hypercube with faulty vertices. The upper bound of longest fault-free cycle of $Q_n - F_v$ is $2^n - 2f_v$ where $f_v = \max\{\mid F_v \cap V_w \mid, \mid F_v \cap V_b \mid\}$, and F_v is the set of vertices of G . In [1, 2, 5, 7], the authors showed that a fault-free cycle of length $2^n - 2f_v$ can be constructed with f_v faulty vertices. This length is optimal when all the faulty vertices have the same color. However, there exist longer cycles when there are faulty vertices with different color.

In [3], Hung et al. studied the adjacent vertices fault tolerance Hamiltonian laceability for hypercube. A set of k pairs of adjacent vertex F_v is consisted of k pairs of adjacent vertices. We can denote F_v as $\{a_1, b_1, a_2, b_2, \dots, a_k, b_k \mid a_i \in V_b, b_i \in V_w \text{ and } (a_i, b_i) \in E \text{ for } 1 \leq i \leq k\}$. A bipartite graph $G = (V_b \cup V_w, E)$ is *f -adjacency k*

*This work was supported in part by the National Science Council of Republic of China under NSC 94-2213-E-212-022-

edges Hamiltonian if $G - F_v$ is k edges Hamiltonian $\forall f$ pairs of adjacent vertex set F_v of G . A bipartite graph $G = (V_b \cup V_w, E)$ is f -adjacency k edges Hamiltonian laceable if $G - F_v$ is k edges Hamiltonian laceable $\forall f$ pairs of adjacent vertex set F_v of G . A bipartite graph $G = (V_b \cup V_w, E)$ is f -adjacency k edges hyper-Hamiltonian laceable if $G - F_v$ is k edges hyper-Hamiltonian laceable $\forall f$ pairs of adjacent vertex set F_v of G . In [3], the authors proved that Q_n is f -adjacency $(n - 2 - f)$ edges Hamiltonian for $n \leq n - 2$, f -adjacency $(n - 2 - f)$ edges Hamiltonian laceable for $f \leq n - 3$, and f -adjacency $(n - 3 - f)$ edges hyper-Hamiltonian laceable for $f \leq n - 3$. They also constructed the fault-free cycle of length $2^n - 2|F_v| + 2$ in $Q_n - F_v$ where F_v is the set of faulty vertices with at least one black vertex and one white vertex. In this paper, we will investigate more general adjacent fault tolerant Hamiltonian laceability. We will construct the fault-free cycle of length $2^n - 2|F_v| + 4$ in $Q_n - F_v$ where F_v is the set of faulty vertices with at least two black vertices and two white vertices.

The rest of this paper is organized as follows. In Section 2, we present some previous results needed for our proof. We show the adjacency fault tolerance for property 2H of hypercube in Section 3. Section 4 proves the adjacency fault tolerance for Hamiltonian laceability of hypercube. The longest fault-free cycle will be constructed in Section 5. Finally, we give some conclusions in Section 6.

2 Preliminaries

In this section, we will present some properties which needed for our main results. In [3], Hung et al. show the following theorem.

Theorem 1 *The graph Q_n is f -adjacency $(n - 2 - f)$ edges Hamiltonian for $0 \leq f \leq (n - 2)$, f -adjacency $(n - 2 - f)$ edges Hamiltonian laceable for $0 \leq f \leq (n - 3)$, and f -adjacency $(n - 3 - f)$ edges hyper-Hamiltonian laceable for $0 \leq f \leq (n - 3)$.*

In [4], Park et al. introduced the concept of property 2H. A bipartite graph $G = (V_b \cup V_w, E)$ has property 2H if it satisfies the following conditions: Let $a_1, a_2 \in V_w$ and $b_1, b_2 \in V_b$, be two pairs of vertices of G . There exist two spanning disjoint paths P_1 and P_2 of G such that (1) P_1 joins a_1 to b_1 , (2) P_2 joins a_2 to b_2 , and (3) every vertex of G is either on P_1 or on P_2 . Let Q_{n-1}^0 and Q_{n-1}^1 be

the $(n - 1)$ -dimensional subgraphs of Q_n . Let x be a vertex of Q_{n-1}^0 , $\phi(x)$ be vertex of Q_{n-1}^1 such that $(x, \phi(x)) \in E$. We denote that A bipartite graph G has k edges property 2H if $G - F_e$ has property 2H where F_e is the set of edges of G and $|F_e| = k$.

The following lemma is showed in [3].

Lemma 1 *The graph $Q_n - F_e$ has property 2H, $F_e \subset E(Q_n)$, $|F_e| = (n - 3)$ for $n \geq 3$.*

We need the following lemma, which is similar to **Lemma 1**, to prove our main results.

Lemma 2 *For $Q_n = (V_b \cup V_w, E)$, $\forall x_0, x_1 \in V_w, \forall y_0, y_1 \in V_b$, there exist two spanning disjoint paths $P(x_0, x_1)$, $P(y_0, y_1)$ in Q_n for $n \geq 4$.*

Proof: By symmetry of hypercube, we can assume $y_0 \in Q_{n-1}^0$ and $y_1 \in Q_{n-1}^1$. We will construct two spanning disjoint paths $P(x_0, x_1)$ and $P(y_0, y_1)$ in the following cases.

Case1: $x_0, x_1 \in Q_{n-1}^0$.

Since Q_{n-1}^0 is hyper Hamiltonian laceable, there exists a Hamiltonian path $P(x_0, x_1)$ of $Q_{n-1}^0 - \{y_0\}$. Since Q_{n-1}^1 is Hamiltonian laceable, there exists a Hamiltonian path $P(\phi(y_0), y_1)$ of Q_{n-1}^1 . Thus, $P(x_0, x_1)$ and $\langle y_0, \phi(y_0) \xrightarrow{P(\phi(y_0), y_1)} y_1 \rangle$ are two spanning disjoint paths of Q_n , as illustrated in Figure 1.(a).

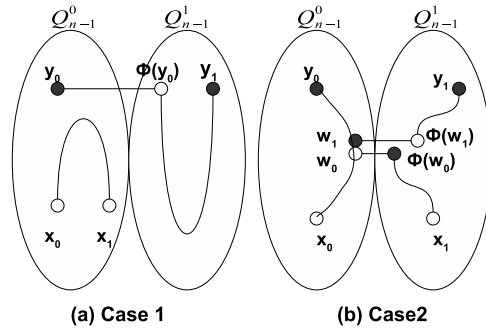


Figure 1: Illustration of **Lemma 2**.

Case2: $x_0 \in Q_{n-1}^0, x_1 \in Q_{n-1}^1$.

Since Q_{n-1}^0 is Hamiltonian laceable, there exists a Hamiltonian path $P(x_0, y_0)$ of Q_{n-1}^0 . We can denote $P(x_0, y_0)$ as $\langle x_0 \xrightarrow{P(x_0, w_0)} w_0, w_1 \xrightarrow{P(w_1, y_0)} y_0 \rangle$, $w_0 \in V_w$, $w_1 \in V_b$ and $\phi(w_0) \neq y_1$, $\phi(w_1) \neq x_1$. Applying **Lemma 1**, we

construct two spanning disjoint paths $P(\phi(w_0), x_1)$ and $P(\phi(w_1), y_1)$ in Q_{n-1}^1 . Thus, $\langle x_0 \xrightarrow{P(x_0, w_0)} w_0, \phi(w_0) \xrightarrow{P(\phi(w_0), x_1)} x_1 \rangle$ and $\langle y_0 \xrightarrow{P(y_0, w_1)} w_1, \phi(w_1) \xrightarrow{P(\phi(w_1), y_1)} y_1 \rangle$ are two spanning disjoint paths of Q_n , as illustrated in Figure 1.(b).

3 The adjacency property 2H of hypercube

In this section, we will investigate the adjacency fault tolerance for property 2H of hypercube. In the follows, we use f_0 and f_1 to denote the numbers of pair of faulty vertices plus faulty edges in Q_{n-1}^0 and Q_{n-1}^1 , respectively. A bipartite graph G has f -adjacency k edges property 2H if $G - F_v$ has k edges property 2H where F_v is the set of f pairs of adjacent vertices.

Lemma 3 *The graph Q_4 is f -adjacency $(1 - f)$ edges property 2H, $f \leq 1$.*

Proof: Applying **Lemma 1**, we can obtain that Q_4 is 1 edge property 2H. In the following, we will show that Q_4 is 1-adjacency property 2H. Let $s_0, s_1 \in V_w$ and $t_0, t_1 \in V_b$ be arbitrary two pairs of vertices in Q_n . Let a, b be two adjacently faulty vertices, $a \in V_b$, $b \in V_w$. By symmetry of hypercube, we can assume $a, b \in Q_{n-1}^0$. We will construct two spanning disjoint paths $P(s_0, t_0)$, $P(s_1, t_1)$ in the following case.

Case 1: $s_0, s_1, t_0, t_1 \in Q_3^0$.

We can assume that $d(s_0, t_0) \leq d(s_1, t_1)$. Since $Q_3^0 - \{a, b\}$ is isomorphic to the graph in Figure 2.(a). There exist three spanning disjoint paths $P(s_0, t_0)$, $P(s_1, x_0)$ and $P(y_0, t_1)$ of $Q_3^0 - \{a, b\}$ for some different color vertices x_0 and y_0 . Since Q_3^1 is Hamiltonian laceable, there exists a Hamiltonian paths $P(\phi(x_0), \phi(y_0))$ of Q_3^1 . Thus, $P(s_0, t_0)$ and $\langle s_1 \xrightarrow{P(s_1, x_0)} x_0, \phi(x_0) \xrightarrow{P(\phi(x_0), \phi(y_0))} \phi(y_0), y_0 \xrightarrow{P(y_0, t_1)} t_1 \rangle$ are two spanning disjoint paths of $Q_4 - \{a, b\}$.

Case 2: one of $s_0, s_1, t_0, t_1 \in Q_3^1$.

Without loss of generality, we can assume that $s_1 \in Q_3^1$. Since $Q_3^0 - \{a, b\}$ is isomorphic to Figure 2.(a), there exist three spanning disjoint paths $P(x_0, s_0)$, $P(y_0, t_0)$, and $P(y_1, t_1)$ of $Q_3^0 - \{a, b\}$ for some $y_1 \in V_w$, $d(x_0, y_0)$ is odd and $s_1 \notin \{\phi(x_0), \phi(y_0)\}$. Since Q_3^1 has

property 2H, there exist two spanning disjoint paths $P(s_1, \phi(y_1))$ and $P(\phi(x_0), \phi(y_0))$. Thus, $\langle s_0 \xrightarrow{P(s_0, x_0)} x_0, \phi(x_0) \xrightarrow{P(\phi(x_0), \phi(y_0))} \phi(y_0), y_0 \xrightarrow{P(y_0, t_0)} t_0 \rangle$ and $\langle s_1 \xrightarrow{P(s_1, \phi(y_1))} \phi(y_1), y_1 \xrightarrow{P(y_1, t_1)} t_1 \rangle$ are two spanning disjoint paths of $Q_4 - \{a, b\}$, as illustrated in Figure 2.(b).

Case 3: $s_0, s_1 \in Q_3^1$ or $t_0, t_1 \in Q_3^1$.

Without loss of generality, we can assume $t_0, t_1 \in Q_3^1$. There exist two spanning disjoint paths $P(s_0, y_0)$ and $P(s_1, x_0)$ of $Q_3^0 - \{a_0, b_0\}$ for some $x_0, y_0 \in V_b$. Since $Q_3^0 - \{a, b\}$ is Hamiltonian, there exist two spanning disjoint paths $P(\phi(x_0), t_1)$ and $P(\phi(y_0), t_0)$ of Q_3^1 . Thus, $\langle s_0 \xrightarrow{P(s_0, y_0)} y_0, \phi(y_0) \xrightarrow{P(\phi(y_0), t_0)} t_0 \rangle$ and $\langle s_1 \xrightarrow{P(s_1, x_0)} x_0, \phi(x_0) \xrightarrow{P(\phi(x_0), t_1)} t_1 \rangle$ are two spanning disjoint paths of $Q_4 - \{a, b\}$, as illustrated in Figure 2.(c).

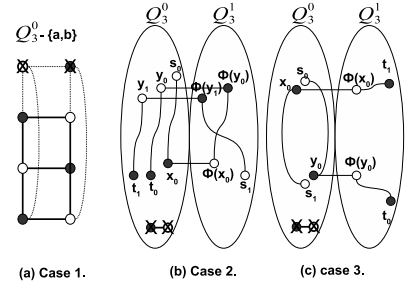


Figure 2: Illustration of **Lemma 3**.

Case 4: $s_0, t_1 \in Q_3^1$ or $s_1, t_0 \in Q_3^1$.

Without loss of generality, we can assume that $s_1, t_0 \in Q_3^1$. When s_1 or t_0 is adjacent to the faulty vertices, there exists a Hamiltonian path $P(s_1, t_0)$ of $Q_3^1 - \{a, b\}$. We can denote $P(s_1, t_0)$ as $\langle s_1 \xrightarrow{P(s_1, x_0)} x_0, y_0 \xrightarrow{P(y_0, t_0)} t_0 \rangle$ where $x_0 \in V_b$, $y_0 \in V_w$ and $\phi(x_0) \neq s_0$, $\phi(y_0) \neq t_1$. Since $Q_3^0 - \{a, b\}$ is Hamiltonian, we can construct two spanning disjoint paths $P(\phi(x_0), t_1)$ and $P(s_0, \phi(y_0))$ of Q_3^1 . Thus, $\langle s_1 \xrightarrow{P(s_1, x_0)} x_0, \phi(x_0) \xrightarrow{P(\phi(x_0), t_1)} t_1 \rangle$ and $\langle s_0 \xrightarrow{P(s_0, \phi(y_0))} \phi(y_0), y_0 \xrightarrow{P(y_0, t_0)} t_0 \rangle$ are two spanning disjoint path of $Q_4 - \{a, b\}$. Suppose s_1 and t_0 are not adjacent to the faulty vertices. The two spanning disjoint path $P(s_0, t_0)$ and $P(s_1, t_1)$ are showed in Figure 3.(a). When s_0, t_0, s_1, t_1 are not as Figure

3.(a), it can be considered as at least 3 vertices in the same subgraphs by symmetry of hypercube.

Case 5: $s_0, t_0 \in Q_3^1$ or $s_1, t_1 \in Q_3^1$.

Without loss of generality, we can assume that $s_1, t_1 \in Q_3^1$. Suppose s_0 or t_0 is adjacent to the faulty vertices. There exists a Hamiltonian path $P(s_0, t_0)$ of $Q_3^0 - \{a, b\}$. There also exists a Hamiltonian path $P(s_1, t_1)$ of Q_3^1 . These two paths are spanning disjoint paths. Suppose s_0 and t_0 are not adjacent to the faulty vertices. We can assume that it can not be consider as at least three vertices in some subgraph Q_3^i by symmetry of hypercube. There exist two spanning disjoint paths $P(s_0, x_0)$ and $P(y_0, t_0)$ of $Q_3^0 - \{a, b\}$ for some $x_0 \in V_b$, $y_0 \in V_w$ and $\phi(x_0) \neq s_1$, $\phi(y_0) \neq t_1$. Applying **Lemma 1**, we can construct two spanning disjoint paths $P(\phi(x_0), \phi(y_0))$, $P(s_1, t_1)$ of Q_3^1 . Thus, $\langle s_0 \xrightarrow{P(s_0, x_0)} x_0, \phi(x_0) \xrightarrow{P(\phi(x_0), \phi(y_0))} \phi(y_0), y_0 \xrightarrow{P(y_0, t_0)} t_0 \rangle$ and $P(s_1, t_1)$ are two spanning disjoint paths of $Q_4 - \{a, b\}$.

Case 6: one of $s_0, t_0, t_1, s_1 \in Q_3^0$.

We can assume that $s_0 \in Q_3^0$. Since $Q_3^0 - \{a, b\}$ is Hamiltonian, we can construct a Hamiltonian path $P(s_0, x_0)$ of $Q_3^0 - \{a, b\}$ for $x_0 \in V_b$, $\phi(x_0) \neq s_1$. We can construct two spanning disjoint paths $P(\phi(x_0), t_0), P(s_1, t_1)$ of Q_3^1 . Thus, $\langle s_0 \xrightarrow{P(s_0, x_0)} x_0, \phi(x_0) \xrightarrow{P(\phi(x_0), t_0)} t_0 \rangle$ and $P(s_1, t_1)$ are two spanning disjoint paths of $Q_4 - \{a, b\}$, as illustrated in Figure 3.(b).

Case 7: $s_0, t_0, t_1, s_1 \in Q_3^1$.

Applying **Lemma 1**, we can construct two spanning paths $P(s_0, t_0)$ and $P(s_1, t_1)$ of $Q^1 - 3 - \{a, b\}$. And there is a Hamiltonian cycle C of $Q_3^0 - \{a, b\}$. Since Q_3^0 and Q_3^1 are 3-regular, there exist two edges (x_0, y_0) of C and $(\phi(x_0), \phi(y_0))$ of $P(s_0, t_0)$ or $P(s_1, t_1)$. We can denote C as $\langle x_0 \xrightarrow{P(x_0, y_0)} y_0, x_0 \rangle$. Without loss of generality, we can assume $(\phi(x_0), \phi(y_0))$ of $P(s_1, t_1)$. Thus, $P(s_0, t_0)$ and $\langle s_1 \xrightarrow{P(s_1, \phi(x_0))} \phi(x_0), x_0 \xrightarrow{P(x_0, y_0)} y_0, \phi(y_0) \xrightarrow{P(\phi(y_0), t_1)} t_1 \rangle$ are two spanning disjoint paths of $Q_4 - \{a, b\}$, as illustrated in Figure 3.(c).

Theorem 2 *The graph Q_n is f -adjacency $(n-3-f)$ edges property $2H$, for $0 \leq f \leq n-3$, $n \geq 3$.*

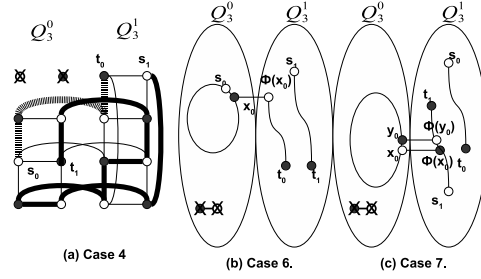


Figure 3: Illustration of **Lemma 3**.

Proof: We will prove this theorem by induction on n . The case for $n = 3$ is true by **Lemma 1**. The case for $n = 4$ is true by **Lemma 3**. In the following, we will discuss the case for $n \geq 5$. By symmetry of hypercube, we can assume every faulty pair $\{a_i, b_i\}$ is either in Q_{n-1}^0 or Q_{n-1}^1 . Let F be a set of faulty vertices and edges, and $F = F_v + F_e$, $|F| = f$. We will construct two spanning disjoint paths $P(s_0, t_0)$, $P(s_1, t_1)$ for any $s_0, s_1 \in V_w$, $t_0, t_1 \in V_b$.

Case 1: $F \subset Q_{n-1}^0$ or $F \subset Q_{n-1}^1$.

Without loss of generality, we can assume $F \subset Q_{n-1}^0$.

Case 1.1: $s_0, s_1, t_0, t_1 \in Q_{n-1}^0$.

Since $Q_n^0 - F$ is Hamiltonian, there exist four spanning disjoint paths $P(s_0, x_0)$, $P(s_1, x_1)$, $P(y_0, t_0)$, $P(y_1, t_1)$ of $Q_{n-1}^1 - F$ for some vertices $x_0, x_1, y_0, y_1 \in Q_{n-1}^0 - F$. Applying **Lemma 1** and **Lemma 2**, we can construct two spanning disjoint paths $P(\phi(x_0), \phi(y_0))$ and $P(\phi(x_1), \phi(y_1))$ of Q_{n-1}^1 . Therefore, $\langle s_0 \xrightarrow{P(s_0, x_0)} x_0, \phi(x_0) \xrightarrow{P(\phi(x_0), \phi(y_0))} \phi(y_0), y_0 \xrightarrow{P(y_0, t_0)} t_0 \rangle$ and $\langle s_1 \xrightarrow{P(s_1, x_1)} x_1, \phi(x_1) \xrightarrow{P(\phi(x_1), \phi(y_1))} \phi(y_1), y_1 \xrightarrow{P(y_1, t_1)} t_1 \rangle$ are two spanning disjoint paths of $Q_n - F$, as illustrated in Figure 4.(a).

Case 1.2: one of $s_0, t_0, t_1, s_1 \in Q_{n-1}^1$.

Without loss of generality, we can assume $s_1 \in Q_{n-1}^1$. Since $Q_{n-1}^0 - F$ is Hamiltonian, there exists three spanning disjoint path $P(s_0, x_0)$, $P(y_0, t_0)$, $P(y_1, t_1)$ of $Q_{n-1}^0 - F$ for some vertices x_0, y_0, y_1 and $s_1 \notin \{\phi(x_0), \phi(y_0), \phi(y_1)\}$. Applying **Lemma 1** and **Lemma 2**, we can construct two spanning disjoint paths $P(s_1, \phi(y_1))$ and $P(\phi(x_0), \phi(y_0))$ of Q_{n-1}^1 .

Therefore, $\langle s_0 \xrightarrow{P(s_0, x_0)} x_0, \phi(x_0) \xrightarrow{P(\phi(x_0), \phi(y_0))} \phi(y_0), y_0 \xrightarrow{P(y_0, t_0)} t_0 \rangle$ and $\langle s_1 \xrightarrow{P(s_1, \phi(y_1))} \phi(y_1), y_1 \xrightarrow{P(y_1, t_1)} t_1 \rangle$ are two spanning disjoint paths of $Q_n - F$, as illustrated in Figure 4.(b).

Case 1.3: $s_0, s_1 \in Q_{n-1}^1$ or $t_0, t_1 \in Q_{n-1}^1$.

Without loss of generality, we can assume $t_0, t_1 \in Q_{n-1}^1$. Since $Q_{n-1}^0 - F$ is Hamiltonian, there exist two spanning disjoint paths $P(s_0, x_0)$ and $P(s_1, x_1)$ of $Q_{n-1}^0 - F$ for some $x_0, x_1 \in V_b$. Applying **Lemma 1**, there exist two spanning disjoint paths $P(\phi(x_0), t_0)$ and $P(\phi(x_1), t_1)$ of Q_{n-1}^1 . Thus, $\langle s_0, \phi(x_0) \xrightarrow{P(s_0, x_0)} x_0, \phi(x_0) \xrightarrow{P(\phi(x_0), t_0)} t_0 \rangle$ and $\langle s_1, \phi(x_1) \xrightarrow{P(s_1, x_1)} x_1, \phi(x_1) \xrightarrow{P(\phi(x_1), t_1)} t_1 \rangle$ are two spanning disjoint paths of $Q_n - F$.

Case 1.4: $s_0, t_1 \in Q_{n-1}^1$ or $t_0, s_1 \in Q_{n-1}^1$.

Without loss of generality, we can assume that $s_1, t_0 \in Q_{n-1}^1$. Since $Q_{n-1}^0 - F$ is Hamiltonian, there exist two spanning disjoint paths $P(s_0, x_0), P(x_1, t_1)$ for some vertices $x_0 \in V_w, x_1 \in V_b$ and $\{\phi(x_0), \phi(x_1)\} \neq \{s_1, t_0\}$. We can assume $\phi(x_0) \neq t_0$. Suppose $\phi(x_1) \neq s_1$. Applying **Lemma 2**, we can construct two spanning disjoint paths $P(s_1, \phi(x_1))$ and $P(\phi(x_0), t_0)$ of Q_{n-1}^1 . Thus, $\langle s_0, \phi(x_0) \xrightarrow{P(s_0, x_0)} x_0, \phi(x_0) \xrightarrow{P(\phi(x_0), t_0)} t_0 \rangle$ and $\langle s_1, \phi(x_1) \xrightarrow{P(s_1, \phi(x_1))} \phi(x_1), x_1 \xrightarrow{P(x_1, t_1)} t_1 \rangle$ are two spanning disjoint paths of $Q_n - F$, as illustrated in Figure 4.(c).

Suppose $\phi(x_1) = s_1$. Since Q_{n-1}^1 is hyper-Hamiltonian laceable, there exists a Hamiltonian path $P(\phi(x_0), t_0)$ of $Q_{n-1}^1 - \{s_1\}$. Thus, $\langle s_0, \phi(x_0) \xrightarrow{P(s_0, x_0)} x_0, \phi(x_0) \xrightarrow{P(\phi(x_0), t_0)} t_0 \rangle$ and $\langle s_1 = \phi(x_1), x_1 \xrightarrow{P(x_1, t_1)} t_1 \rangle$ are two spanning disjoint paths of $Q_n - F$.

Case 1.5: $s_0, t_0 \in Q_{n-1}^1$ or $s_1, t_1 \in Q_{n-1}^1$.

Without loss of generality, we can assume that $s_1, t_1 \in Q_{n-1}^1$. Suppose that (s_0, t_0) is an edge of the Hamiltonian cycle of $Q_{n-1}^0 - F$. Thus, the Hamiltonian path $P(s_0, t_0)$ of $Q_{n-1}^0 - F$ and the Hamiltonian path $P(s_1, t_1)$ of Q_{n-1}^1 are two spanning disjoint paths of $Q_n - F$. Suppose that (s_0, t_0) is not an edge of the Hamiltonian cycle of $Q_{n-1}^0 - F$. There exist two spanning disjoint paths $P(s_0, t_0)$ and $P(x_1, y_1)$ for some $x_1 \in V_b$ and $y_1 \in V_w$ and $\{\phi(x_1), \phi(y_1)\} \neq \{s_1, t_1\}$. We can assume $\phi(y_1) \neq t_1$. When $\phi(x_1) \neq s_0$, there exist two spanning disjoint paths $P(s_1, \phi(y_1))$

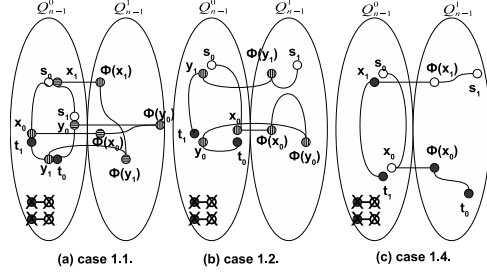


Figure 4: Illustration of **Theorem 2**.

and $P(\phi(x_1), t_1)$ of Q_{n-1}^1 . Thus, $P(s_0, t_0)$ and $\langle s_1, \phi(y_1) \xrightarrow{P(s_1, \phi(y_1))} \phi(y_1), y_1 \xrightarrow{P(y_1, x_1)} x_1, \phi(x_1) \xrightarrow{P(\phi(x_1), t_1)} t_1 \rangle$ are two spanning disjoint paths of $Q_n - F$, as illustrated in Figure 5.(a). When $\phi(x_1) = s_1$, there exists a Hamiltonian path $P(\phi(y_1), t_1)$ of $Q_{n-1}^1 - F$. Thus, $P(s_0, t_0)$ and $\langle s_1 = \phi(x_1), x_1 \xrightarrow{P(x_1, y_1)} y_1, \phi(y_1) \xrightarrow{P(\phi(y_1), t_1)} t_1 \rangle$ are two spanning disjoint paths of $Q_n - F$.

Case 1.6: one of $s_0, t_0, s_1, t_1 \in Q_{n-1}^0$.

Without loss of generality, we can assume that $s_0 \in Q_{n-1}^0$. Since $Q_{n-1}^0 - F$ is Hamiltonian, there exists a Hamiltonian path $P(s_0, x_0)$ of $Q_{n-1}^0 - F$ for some $x_0 \in V_b$ and $\phi(x_0) \neq s_1$. Applying **Lemma 1**, we can construct two spanning disjoint paths $P(\phi(x_0), t_0), P(s_1, t_1)$ of Q_{n-1}^1 . Thus, $\langle s_0, \phi(x_0) \xrightarrow{P(s_0, x_0)} x_0, \phi(x_0) \xrightarrow{P(\phi(x_0), t_0)} t_0 \rangle$ and $P(s_1, t_1)$ are two spanning disjoint paths of Q_n as illustrated in Figure 5.(b).

Case 1.7: $s_0, t_0, t_1, s_1 \in Q_{n-1}^1$.

Let s_0' be the symmetrical vertex of s_0 , and $s_0' \in Q_{n-1}^0$. Suppose $s_0' = F$. Let $\phi(x_0)$ be a neighbor of s_0 such that $x_0 \notin F$. Since $Q_{n-1}^0 - F$ is Hamiltonian, there exists a Hamiltonian path $P(x_0, y_0)$ for some vertex $y_0 \in V_b$ and $\phi(y_0) \neq s_1$. Since Q_{n-1}^1 is 1-adjacency property 2H, there exists two spanning disjoint paths $P(s_1, t_1)$ and $P(\phi(y_0), t_0)$ of $Q_{n-1}^1 - \{s_0, \phi(x_0)\}$. Thus, $\langle s_0, \phi(x_0) \xrightarrow{P(x_0, y_0)} y_0, \phi(y_0) \xrightarrow{P(\phi(y_0), t_0)} t_0 \rangle$ and $P(s_1, t_1)$ are two spanning disjoint paths of $Q_n - F$ as illustrated in Figure 5.(c). Suppose $s_0' \neq F$. Since $Q_{n-1}^0 - F$ is Hamiltonian, there exist a Hamiltonian path $P(s_0', x)$

of $Q_{n-1}^0 - F$ for some $x \in V_w$ and $\phi(x) \neq t_1$. Suppose $\phi(x) = t_0$. Since $Q_{n-1}^1 - \{s_0, t_0\}$ is Hamiltonian laceable, there exist a Hamiltonian path $P(s_1, t_1)$ of $Q_{n-1}^1 - \{s_0, t_0\}$. Thus, $\langle s_0, s_0' \xrightarrow{P(s_0', x)} x, \phi(x) = t_0 \rangle$ and $P(s_1, t_1)$ are two spanning disjoint paths of $Q_n - F$. Suppose $\phi(x) \neq t_0$, and $(s_0, \phi(x))$ is an edge of $E(Q_{n-1}^0)$. Let w is a neighbor of $\phi(x)$, and $w \neq s_1$. Since $Q_{n-1}^1 - \{s_0, \phi(x)\}$ is property 2H, there exists two spanning disjoint paths $P(s_1, t_1)$ and $P(\phi(w), t_0)$ of $Q_{n-1}^1 - \{s_0, \phi(x)\}$. Thus, $\langle s_0, s_0' \xrightarrow{P(s_0', x)} x, \phi(x), w \xrightarrow{P(w, t_0)} t_0 \rangle$ and $P(s_1, t_1)$ are two spanning disjoint paths of $Q_n - F$.

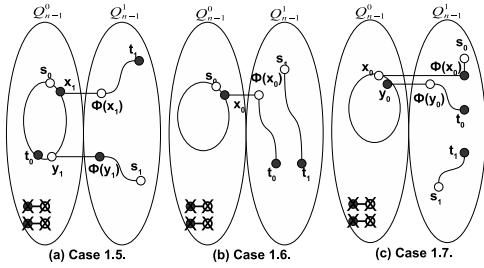


Figure 5: Illustration of Theorem 2.

Case 2: $1 \leq f_0, f_1 \leq n - 4$.

Case 2.1: $s_0, s_1, t_0, t_1 \in Q_{n-1}^0$.

Since Q_{n-1}^0 is f -adjacency $(n - 4 - f)$ edges property 2H, we can construct two spanning disjoint paths $P(s_0, t_0)$ and $P(s_1, t_1)$ in $Q_{n-1}^0 - F$. We can assume that the length of $P(s_0, t_0)$ is longer than the length of $P(s_1, t_1)$. Let (x_0, y_0) be an edge of $P(s_0, t_0)$ such that $\phi(x_0), \phi(y_0)$ are fault-free. Applying **Theorem 1**, we can construct a Hamiltonian path $P(\phi(x_0), \phi(y_0))$ in $Q_{n-1}^1 - F$. Thus, $\langle s_0, \xrightarrow{P(s_0, x_0)} x_0, \phi(x_0) \xrightarrow{P(\phi(x_0), \phi(y_0))} \phi(y_0), y_0 \xrightarrow{P(y_0, t_0)} t_0 \rangle$ and $P(s_1, t_1)$ are two spanning disjoint paths of $Q_n - F$, as illustrated in Figure 6.(a).

Case 2.2: one of $s_0, t_0, t_1, s_1 \in Q_{n-1}^1$.

Without loss of generality, we assume $t_1 \in Q_{n-1}^1$. Since Q_{n-1}^0 is f -adjacency property 2H for $0 \leq f \leq n - 4$, there exist two spanning disjoint paths $P(s_0, t_0)$ and $P(s_1, x)$ in $Q_{n-1}^0 - F$ where $x \in V_b$ and $\phi(x)$ is fault-free. Since Q_{n-1}^1 is f -adjacency Hamiltonian laceable, there exists a Hamiltonian path

$P(\phi(x), t_1)$ of $Q_{n-1}^1 - F^1$. Thus, $P(s_0, t_0)$ and $\langle s_1 \xrightarrow{P(s_1, x)} x, \phi(x) \xrightarrow{P(\phi(x), t_1)} t_1 \rangle$ are two spanning disjoint paths of $Q_n - F$, as illustrated in Figure 6.(b).

Case 2.3: two of $s_0, s_1, t_0, t_1 \in Q_{n-1}^1$.

Without loss of generality, we can assume that $s_0, t_1 \in Q_{n-1}^1$. Let $x_0, y_0, \phi(x_0), \phi(y_0)$ be fault-free vertices where the color of x_0 (resp. y_0) is different with s_0 (resp. t_1). Since Q_{n-1} is f -adjacency property 2H for $0 \leq f \leq n - 4$, there exist two spanning disjoint paths $P(s_0, x_0)$ and $P(y_0, t_1)$ of $Q_{n-1}^0 - F$ and two spanning disjoint paths $P(s_1, \phi(y_0))$ and $P(\phi(x_0), t_0)$ of $Q_{n-1}^1 - F^1$. Thus, $\langle s_0, \xrightarrow{P(s_0, x_0)} x_0, \phi(x_0) \xrightarrow{P(\phi(x_0), t_0)} t_0 \rangle$ and $\langle s_1 \xrightarrow{P(s_1, \phi(y_0))} \phi(y_0), y_0 \xrightarrow{P(y_0, t_1)} t_1 \rangle$ are two spanning disjoint paths of $Q_n - F$, as illustrated in Figure 6.(c).

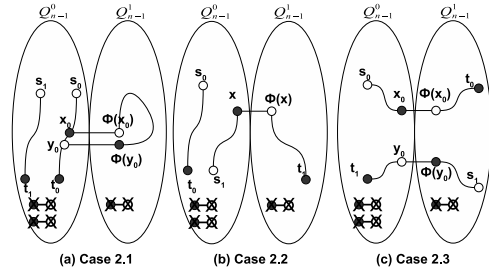


Figure 6: Illustration of Theorem 2.

4 The adjacency Hamiltonian laceability of hypercube

In this section, we will show some adjacency fault tolerance Hamiltonian laceability for hypercube.

Lemma 4 Let $Q_4 = (V_b \cup V_w, E)$. For arbitrary $a \in V_b, b \in V_w, e \in E$, the graph $Q_4 - \{a, b, e\}$ is Hamiltonian laceable.

Proof: By symmetry of hypercube, we can assume that $a, b \in Q_3^0$. We will construct a Hamiltonian path between s and t for $s \in V_w, t \in V_b$ in following cases.

Case 1: $s, t \in Q_3^0$.

Case 1.1: $e \in Q_3^0$.

Since Q_3^0 is 1-edge Hamiltonian laceable, there exist a Hamiltonian path $P(s, t)$ of $Q_3^0 -$

$\{e\}$. Suppose that (a, b) is an edge of $P(s, t)$. We can denote $P(s, t)$ as $\langle s \xrightarrow{P(s,x)} x, a, b, y \xrightarrow{P(y,t)} t \rangle$. Since Q_3^1 is Hamiltonian laceable, there exist a Hamiltonian path $P(\phi(x), \phi(y))$ of Q_3^1 . Thus, $\langle s \xrightarrow{P(s,x)} x, \phi(x) \xrightarrow{P(\phi(x),\phi(y))} \phi(y), y \xrightarrow{P(y,t)} t \rangle$ is a Hamiltonian path of $Q_4 - \{a, b, e\}$. Suppose that (a, b) is not an edge of $P(s, t)$. We can denote $P(s, t)$ as $\langle s \xrightarrow{P(s,x_0)} x_0, a, x_1 \xrightarrow{P(x_1,y_1)} y_1, b, y_0 \xrightarrow{P(y_0,t)} t \rangle$. Applying **Lemma 1**, we can construct two spanning disjoint paths $P(\phi(x_1), \phi(y_0))$ and $P(\phi(x_0), \phi(y_1))$. Thus, $\langle s \xrightarrow{P(s,x_0)} x_0, \phi(x_0) \xrightarrow{P(\phi(x_0),\phi(y_1))} \phi(y_1), y_1 \xrightarrow{P(y_1,x_1)} x_1, \phi(x_1) \xrightarrow{P(\phi(x_1),\phi(y_0))} \phi(y_0), y_0 \xrightarrow{P(y_0,t)} t \rangle$ is a Hamiltonian path of $Q_4 - \{a, b, e\}$ as illustrated in Figure 7.(a).

Case 1.2: $e \notin Q_3^0$.

Since $Q_3^0 - \{a_0, b_0\}$ is Hamiltonian, there exist two spanning disjoint paths $P(s, x)$ and $P(y, t)$ of Q_3^0 for some x, y where $d(x, y)$ is odd and $e \notin \{(x, \phi(x)), (y, \phi(y))\}$. There exists a Hamiltonian path $P(\phi(x), \phi(y))$ of Q_3^1 . Thus, $\langle s \xrightarrow{P(s,x)} x, \phi(x) \xrightarrow{P(\phi(x),\phi(y))} \phi(y), y \xrightarrow{P(y,t)} t \rangle$ is a Hamiltonian path of $Q_4 - \{a, b, e\}$ as illustrated in Figure 7.(b).

Case 2: $s \in Q_3^0$ and $t \in Q_3^1$.

Case 2.1: $e \in Q_3^0$.

Since Q_3^0 is 1-edge Hamiltonian laceable, there exists a Hamiltonian path $P(s, a)$ of $Q_3^0 - \{e\}$. Suppose that (a, b) is an edge of $P(s, a)$. We denote $P(s, a)$ as $\langle s \xrightarrow{P(s,x)} x, b, a \rangle$. There exists a Hamiltonian path $P(\phi(x), t)$ of Q_3^1 . Thus, $\langle s \xrightarrow{P(s,x)} x, \phi(x) \xrightarrow{P(\phi(x),t)} t \rangle$ is a Hamiltonian path of $Q_4 - \{a, b, e\}$. Suppose that (a, b) is not an edge of $P(s, a)$. We denote $P(s, a)$ as $\langle s \xrightarrow{P(s,x)} x, b, y \xrightarrow{P(y,z)} z, a \rangle$. Suppose $\phi(z) \neq t$. Since Q_3^1 has property 2H, there exist two spanning disjoint paths $P(\phi(x), \phi(z))$ and $P(\phi(y), t)$ of Q_3^1 . Thus, $\langle s \xrightarrow{P(s,x)} x, \phi(x) \xrightarrow{P(\phi(x),\phi(z))} \phi(z), z \xrightarrow{P(z,y)} y, \phi(y) \xrightarrow{P(\phi(y),t)} t \rangle$ is a Hamiltonian path of $Q_4 - \{a, b, e\}$, as illustrated in Figure 7.(c). Suppose $\phi(z) = t$. Since Q_3^1 is hyper Hamiltonian laceable, there exists a Hamiltonian path $P(\phi(x), \phi(y))$ of $Q_3^1 - \{t\}$. Thus, $\langle s \xrightarrow{P(s,x)} x, \phi(x) \xrightarrow{P(\phi(x),\phi(y))} \phi(y), y \xrightarrow{P(y,z)} z, \phi(z) = t \rangle$ is a Hamiltonian path of $Q_4 - \{a, b, e\}$.

Case 2.2: $e \notin Q_3^0$.

Since $Q_3^0 - \{a, b\}$ is Hamiltonian, there exists a Hamiltonian cycle $\langle s \xrightarrow{P(s,x)} x, s \rangle$ of $Q_3^0 - \{a, b\}$ where $(x, \phi(x)) \neq e$. Since Q_3^1 is 1-edge Hamiltonian laceable, there exists a Hamiltonian path $P(\phi(x), t)$ of $Q_3^1 - \{e\}$. Thus, $\langle s \xrightarrow{P(s,x)} x, \phi(x) \xrightarrow{P(\phi(x),t)} t \rangle$ is a Hamiltonian path of $Q_4 - \{a, b, e\}$, as illustrated in Figure 7.(d).

Case 3: $s, t \in Q_3^1$.

Case 3.1: $e \in Q_3^0$.

Let $e = (u, v)$ for $u \in V_b, v \in V_w$. Since $Q_3^0 - \{a, b\}$ is Hamiltonian, there exists a Hamiltonian cycle C_1 of $Q_3^0 - \{a, b\}$. Suppose that e is not an edge of C_1 . We denote C_1 as $\langle x \xrightarrow{P(x,y)} y, x \rangle$ for $x \in V_b, y \in V_w$ and $\phi(x) \neq t, \phi(y) \neq s$. Since Q_3^1 has property 2H, there exist two spanning disjoint paths $P(s, \phi(x))$ and $P(\phi(y), t)$ of Q_3^1 . Thus, $\langle s \xrightarrow{P(s,\phi(x))} \phi(x), x \xrightarrow{P(x,y)} y, \phi(y) \xrightarrow{P(\phi(y),t)} t \rangle$ is a Hamiltonian path of $Q_4 - \{a, b, e\}$. Suppose that $e = (u, v)$ is an edge of C_1 . Hence, we can denote C_1 as $\langle v \xrightarrow{P(v,u)} u, v \rangle$. If $s \neq \phi(u)$ and $t \neq \phi(v)$, there exist two spanning disjoint paths $P(s, \phi(v))$ and $P(\phi(u), t)$ of Q_3^1 . Thus, $\langle s \xrightarrow{P(s,\phi(v))} \phi(v), v \xrightarrow{P(v,u)} u, \phi(u) \xrightarrow{P(\phi(u),t)} t \rangle$ is a Hamiltonian path of $Q_4 - \{a, b, e\}$ as illustrated in Figure 7.(e). Suppose $|\{s, t\} \cap \{\phi(u), \phi(v)\}| = 1$. We can assume that $s = \phi(u)$ and $t \neq \phi(v)$. Since Q_3 is hyper-Hamiltonian laceable, there exists a Hamiltonian path $P(\phi(v), t)$ of $Q_3^1 - \{s\}$. Thus, $\langle s = \phi(u), u \xrightarrow{P(u,v)} v, \phi(v) \xrightarrow{P(\phi(v),t)} t \rangle$ is a Hamiltonian path of $Q_4 - \{a, b, e\}$. Suppose $s = \phi(u)$ and $t = \phi(v)$. There exists a Hamiltonian cycle C_2 of $Q_3^1 - \{s, t\}$. Since Q_3 is 3-regular, the $P(u, v)$ and C_2 can be denoted as $\langle u \xrightarrow{P(u,x)} x, y \xrightarrow{P(y,v)} v \rangle$ and $\langle \phi(x) \xrightarrow{P(\phi(x),\phi(y))} \phi(y), \phi(x) \rangle$ respective. Thus, $\langle s = \phi(u), u \xrightarrow{P(u,x)} x, \phi(x) \xrightarrow{P(\phi(x),\phi(y))} \phi(y), y \xrightarrow{P(y,v)} v, \phi(v) = t \rangle$ is a Hamiltonian path of $Q_4 - \{a, b, e\}$.

Case 3.2: $e \notin Q_3^0$.

There exists a Hamiltonian cycle C_1 of $Q_3^0 - \{a, b\}$ and a Hamiltonian path $P(s, t)$ of $Q_3^1 - \{e\}$. Since Q_3 is 3-regular, the C_1 and $P(s, t)$ can be denoted as $\langle x \xrightarrow{P(x,y)} y, x \rangle$ and $\langle s \xrightarrow{P(s,\phi(x))} \phi(x), \phi(y) \xrightarrow{P(\phi(y),t)} t \rangle$ where

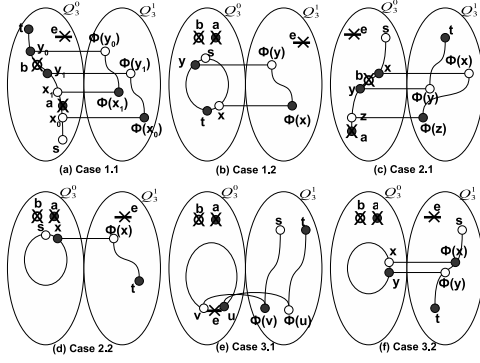


Figure 7: Illustration of Lemma 4.

$(x, \phi(x)), (y, \phi(y)) \neq e$, respectively. Thus, $\langle s \xrightarrow{P(s, \phi(x))} \phi(x), x \xrightarrow{P(x, y)} y, \phi(y) \xrightarrow{P(\phi(y), t)} t \rangle$ is a Hamiltonian path of $Q_n - \{a, b, e\}$ as illustrated in Figure 7.(f).

Theorem 3 Let $Q_n = \{V_b \cup V_w, E\}$. For $a \in V_b, b \in V_w$, the graph $Q_n - \{a, b\}$ is f -adjacency $(n - f - 3)$ edges Hamiltonian laceable for $0 \leq f \leq n - 4, n \geq 4$.

Proof: We will prove this theorem by induction on n . The case for $n=4$ is true by Lemma 4. By symmetry of hypercube, we can assume every pair of adjacently faulty vertices a_i, b_i and faulty edges are either in Q_{n-1}^0 or Q_{n-1}^1 . We denote F is a set of faulty vertices and edges, and $|F| = |F_v| + |F_e|$. We also denote that the set of faulty vertices and edges of Q_{n-1}^0 is F^0 and the set of faulty vertices and edges of Q_{n-1}^1 is F^1 . We will construct a Hamiltonian paths $P(s, t)$ of $Q_n - F - \{a, b\}$ for $n \geq 5$ in the following cases.

Case 1: $F \subset Q_{n-1}^0$ or $F \subset Q_{n-1}^1$.
We assume that $F \subset Q_{n-1}^0$.

Case 1.1: $a, b \in Q_{n-1}^0$.

Case 1.1.1: $s, t \in Q_{n-1}^0$.

Applying Theorem 2, there exist two spanning disjoint paths $P(s, a)$ and $P(b, t)$ for $Q_{n-1}^0 - F$. We denote $P(s, a)$ as $\langle s \xrightarrow{P(s, y)} y, a \rangle$ and $P(b, t)$ as $\langle b, x \xrightarrow{P(x, t)} t \rangle, x \in V_b, y \in V_w$. Since Q_{n-1}^1 is Hamiltonian laceable, there exists a Hamiltonian path $P(\phi(y), \phi(x))$. Thus, $\langle s \xrightarrow{P(s, y)} y, \phi(y) \xrightarrow{P(\phi(y), \phi(x))} \phi(x), x \xrightarrow{P(x, t)} t \rangle$ is a Hamiltonian path of $Q_n - F - \{a, b\}$ as illustrated in Figure 8.(a).

Case 1.1.2: $s \in Q_{n-1}^0, t \in Q_{n-1}^1$.

Applying Theorem 1, we can construct a

Hamiltonian path $\langle s \xrightarrow{P(s, x)} x, b \rangle$ for $Q_{n-1}^0 - F - \{a\}$, $x \in V_b$. Since Q_{n-1}^1 is Hamiltonian laceable, there exists a Hamiltonian paths $P(\phi(x), t)$ for Q_{n-1}^1 . Thus, $\langle s \xrightarrow{P(s, x)} x, \phi(x) \xrightarrow{P(\phi(x), t)} t \rangle$ is a Hamiltonian path of $Q_n - F - \{a, b\}$, as illustrated in Figure 8.(b).

Case 1.1.3: $s, t \in Q_{n-1}^1$.

Applying Theorem 1, there exist a Hamiltonian path $P(y, b)$ of $Q_{n-1}^0 - F - \{a\}$ for $\phi(y) \neq t$. We denote $P(y, b)$ as $\langle y \xrightarrow{P(y, x)} x, b \rangle$ and $x \in V_b$. Suppose $\phi(x) = s$. Since Q_{n-1}^1 is hyper-Hamiltonian laceable, there exists a Hamiltonian path $P(\phi(y), t)$ in Q_{n-1}^1 . Thus, $\langle s, x \xrightarrow{P(x, y)} y, \phi(y) \xrightarrow{P(\phi(y), t)} t \rangle$ is a Hamiltonian path of $Q_n - F - \{a, b\}$. Suppose $\phi(x) \neq s$. Applying Lemma 1, there exist two spanning disjoint paths $P(\phi(x), t)$ and $P(s, \phi(y))$ of Q_{n-1}^1 . Thus, $\langle s \xrightarrow{P(s, \phi(y))} \phi(y), y \xrightarrow{P(y, x)} x, \phi(x) \xrightarrow{P(\phi(x), t)} t \rangle$ is a Hamiltonian path of $Q_n - F - \{a, b\}$, as illustrated in Figure 8.(c).

Case 1.2: $b \in Q_{n-1}^0, a \in Q_{n-1}^1$ or $b \in Q_{n-1}^1, a \in Q_{n-1}^0$.

Without loss of generality, we assume that $b \in Q_{n-1}^0$, and $a \in Q_{n-1}^1$.

Case 1.2.1: $s, t \in Q_{n-1}^0$ or $s, t \in Q_{n-1}^1$.

We assume $s, t \in Q_{n-1}^0$. Since $Q_{n-1}^0 - F$ is Hamiltonian laceable, there exists a Hamiltonian path $P(s, t)$. We denote $P(s, t)$ as $\langle s \xrightarrow{P(s, x_1)} x_1, b, x_0 \xrightarrow{P(x_0, t)} t \rangle$. Since Q_{n-1}^1 is hyper-Hamiltonian laceable, there exists a Hamiltonian path $P(\phi(x_1), \phi(x_0))$ of $Q_{n-1}^1 - \{a\}$. Thus, $\langle s \xrightarrow{P(s, x_1)} x_1, \phi(x_1) \xrightarrow{P(\phi(x_1), \phi(x_0))} \phi(x_0), x_0 \xrightarrow{P(x_0, t)} t \rangle$ is a Hamiltonian path of $Q_n - F - \{a, b\}$, as illustrated in Figure 8.(d)..

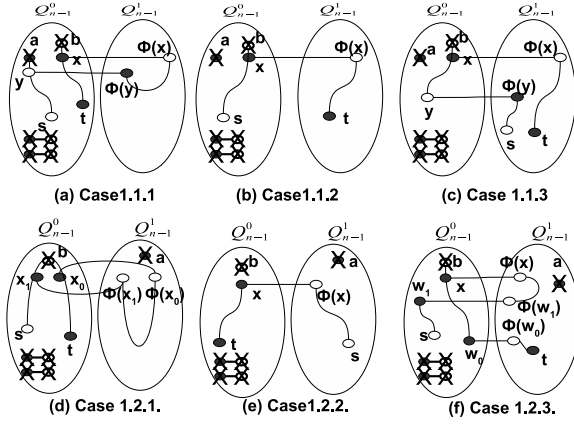
Case 1.2.2: $s \in Q_{n-1}^1, t \in Q_{n-1}^0$.

Since $Q_{n-1}^0 - F$ is hyper-Hamiltonian laceable, there exists a Hamiltonian path $P(x, t)$ of Q_{n-1}^0 . Since Q_{n-1}^1 is hyper-Hamiltonian laceable, there exists a Hamiltonian path $P(s, \phi(x))$. Thus, $\langle s \xrightarrow{P(s, \phi(x))} \phi(x), x \xrightarrow{P(x, t)} t \rangle$ is a Hamiltonian path of $Q_n - F - \{a, b\}$ as illustrated in Figure 8.(e).

Case 1.2.3: $s \in Q_{n-1}^0, t \in Q_{n-1}^1$.

Let $\phi(w_0)$ be a neighbor of t such that w_0 is fault-free and $d(b, w_0) > 1$. Applying Theorem 2, we can obtain two spanning disjoint paths $P(s, w_1)$ and $P(\phi(w_0), b)$ in $Q_{n-1}^0 -$

F . We denote $P(w_0, b)$ as $\langle w_0 \xrightarrow{P(w_0, x)} x, b \rangle$, $x \in V_b$. Applying **Theorem 1**, we can obtain a Hamiltonian path $P(\phi(x_0), \phi(w_1))$ of $Q_{n-1}^1 - \{a, t, w_0\}$. Thus, $\langle s \xrightarrow{P(s, w_1)} w_1, \phi(w_1) \xrightarrow{P(\phi(w_1), \phi(x))} \phi(x), x \xrightarrow{P(x, \phi(w_0))} \phi(w_0), w_0 \xrightarrow{P(w_0, t)} t \rangle$ is a Hamiltonian path of $Q_n - F - \{a, b\}$, as illustrated in Figure 8.(f). Suppose $\phi(b) = t$. Applying **Theorem 2**, we can obtain two spanning disjoint paths $P(s, w_1)$ and $\langle w_0 \xrightarrow{P(w_0, x)} x, b \rangle$ of $Q_{n-1}^0 - F$ for $w_0, w_1 \in V_b$. Applying **Theorem 1**, we can construct a Hamiltonian path $P(\phi(w_1), \phi(w_0))$ of $Q_{n-1}^1 - \{t, \phi(x), a\}$. Thus, $\langle s \xrightarrow{P(s, w_1)} w_1, \phi(w_1) \xrightarrow{P(\phi(w_1), \phi(w_0))} \phi(w_0), w_0 \xrightarrow{P(w_0, x)} x, \phi(x), t \rangle$ is a Hamiltonian path of $Q_n - F - \{a, b\}$.


 Figure 8: Illustration of **Theorem 3**.

Case 1.3: $a, b \in Q_{n-1}^1$.

Case 1.3.1: $s, t \in Q_{n-1}^0$.

Since $Q_{n-1}^0 - F$ is property 2H, there exist two spanning disjoint paths $P(s, x)$, $P(t, y)$, and $\phi(x) \neq b$, $\phi(y) \neq a$, $x \in V_b$, $y \in V_w$. By induction, $Q_{n-1}^1 - \{a, b\}$ is Hamiltonian laceable, there exist a Hamiltonian path $P(\phi(x), \phi(y))$. Thus, $\langle s \xrightarrow{P(s, x)} x, \phi(x) \xrightarrow{P(\phi(x), \phi(y))} \phi(y), y \xrightarrow{P(y, t)} t \rangle$ is a Hamiltonian path of $Q_n - F - \{a, b\}$.

Case 1.3.2: $s \in Q_{n-1}^0$, $t \in Q_{n-1}^1$ or $s \in Q_{n-1}^1$, $t \in Q_{n-1}^0$.

We can assume that $s \in Q_{n-1}^0$ and $t \in Q_{n-1}^1$. Since $Q_{n-1}^0 - F^0$ is Hamiltonian laceable, there exists a Hamiltonian path

$P(s, x)$, and $x \in V_b$, $\phi(x) \neq b$. Since $Q_{n-1}^1 - \{a, b\}$ is Hamiltonian laceable, there exists a Hamiltonian path $P(\phi(x), t)$. Thus, $\langle s \xrightarrow{P(s, x)} x, \phi(x) \xrightarrow{P(\phi(x), t)} t \rangle$ is a Hamiltonian path of $Q_n - F - \{a, b\}$.

Case 1.3.3: $s, t \in Q_{n-1}^1$.

Since $Q_{n-1}^1 - \{a, b\}$ is Hamiltonian laceable, there exist a Hamiltonian path $\langle s \xrightarrow{P(s, \phi(x))} \phi(x), \phi(y) \xrightarrow{P(\phi(y), t)} t \rangle$ for $x, y \notin F$. Since Q_{n-1}^0 and Q_{n-1}^1 are $(n-1)$ -regular, there exist two vertices x, y in Q_{n-1}^0 and $\phi(x_0), \phi(y_0)$ are in Q_{n-1}^1 of $P(s, t)$, and $x, y \notin F$. Since $Q_{n-1}^0 - F$ is Hamiltonian laceable, there exist a Hamiltonian path $P(x, y)$. Thus, $\langle s \xrightarrow{P(s, \phi(x))} \phi(x), x \xrightarrow{P(x, y)} y, \phi(y) \xrightarrow{P(\phi(y), t)} t \rangle$ is a Hamiltonian path of $Q_n - F - \{a, b\}$.

Case 2: $1 \leq f_0, f_1 \leq n - 5$.

Case 2.1: $a, b \in Q_{n-1}^0$ or $a, b \in Q_{n-1}^1$.

We assume that $a, b \in Q_{n-1}^0$.

Case 2.1.1: $s, t \in Q_{n-1}^0$.

Since $Q_{n-1}^0 - \{a, b\}$ is f -adjacency $(n - 4 - f)$ edges Hamiltonian laceable, there exist a Hamiltonian path $\langle s \xrightarrow{P(s, x)} x, y \xrightarrow{P(y, t)} t \rangle$ of $Q_{n-1}^0 - F^0 - \{a, b\}$ for $\phi(x), \phi(y) \notin F$. Applying **Theorem 1**, we can construct a Hamiltonian path $P(\phi(x), \phi(y))$ of $Q_{n-1}^1 - F^1$. Thus, $\langle s \xrightarrow{P(s, x)} x, \phi(x) \xrightarrow{P(\phi(x), \phi(y))} \phi(y), y \xrightarrow{P(y, t)} t \rangle$ is a Hamiltonian path of $Q_n - F - \{a, b\}$, as illustrated in Figure 9.(a).

Case 2.1.2: $s \in Q_{n-1}^0$, $t \in Q_{n-1}^1$ or $t \in Q_{n-1}^0$, $s \in Q_{n-1}^1$.

Without loss of generality, we can assume that $s \in Q_{n-1}^0$ and $t \in Q_{n-1}^1$. Let $x \in Q_{n-1}^0$ and $\phi(x) \in Q_{n-1}^1$ be fault-free vertices. There exist a Hamiltonian path $P(s, x)$ of $Q_{n-1}^0 - F^0 - \{a, b\}$ and a Hamiltonian path $P(\phi(x), t)$ of $Q_{n-1}^1 - F^1$. Thus, $\langle s \xrightarrow{P(s, x)} x, \phi(x) \xrightarrow{P(\phi(x), t)} t \rangle$ is a Hamiltonian path of $Q_n - F - \{a, b\}$ as illustrated in Figure 9.(b).

Case 2.1.3: $\{s, t\} \in Q_{n-1}^1$.

Let $x, y \in Q_{n-1}^0$ and $\phi(x), \phi(y) \in Q_{n-1}^1$ be fault-free vertices for $\{\phi(x), \phi(y)\} \cap \{s, t\} = \emptyset$. Applying **Theorem 2**, we can obtain two spanning disjoint paths $P(s, \phi(y))$ and $P(\phi(x), t)$ of $Q_{n-1}^1 - F^1$. Since $Q_{n-1}^0 - \{a, b\}$ is f -adjacency $(n - 4 - f)$ edges Hamiltonian laceable, there exists a Hamiltonian

path $P(y, x)$ in $Q_{n-1}^0 - F^0 - \{a, b\}$. Thus, $\langle s \xrightarrow{P(s, \phi(y))} \phi(y), y \xrightarrow{P(y, x)} x, \phi(x) \xrightarrow{P(\phi(x), t)} t \rangle$ is a Hamiltonian path of $Q_n - F - \{a, b\}$ as illustrated in Figure 9.(c).

Case 2.2: $a \in Q_{n-1}^1$, $b \in Q_{n-1}^0$ or $a \in Q_{n-1}^0$, $b \in Q_{n-1}^1$.

We assume that $a \in Q_{n-1}^1$ and $b \in Q_{n-1}^0$.

Case 2.2.1: $s, t \in Q_{n-1}^0$.

Applying **Theorem 2**, there exist two spanning disjoint paths $P(s, x_0)$ and $P(t, b)$.

We denote $P(b, t)$ as $\langle b, x_1 \xrightarrow{P(x_1, t)} t \rangle$ and $\phi(x_0), \phi(x_1) \notin F^1$. Applying **Theorem 1**, we can construct a Hamiltonian path $P(\phi(x_0), \phi(x_1))$ of $Q_{n-1}^1 - F^1 - \{a\}$. Thus, $\langle s \xrightarrow{P(s, x_0)} x_0, \phi(x_0) \xrightarrow{P(\phi(x_0), \phi(x_1))} \phi(x_1), x_1 \xrightarrow{P(x_1, t)} t \rangle$ is a Hamiltonian path of $Q_n - F - \{a, b\}$ as illustrated in Figure 9.(d).

Case 2.2.2: $s \in Q_{n-1}^0$ and $t \in Q_{n-1}^1$.

Suppose $\phi(b) \neq t$. Let $\phi(y)$ be a neighbor of t such that y is fault-free and $d(b, y) > 1$. Applying **Theorem 2**, we obtain two spanning disjoint paths $P(s, x_0)$, $P(y_0, b)$. We denote $P(y, b)$ as $\langle y \xrightarrow{P(y, x_1)} x_1, b \rangle$, and $\phi(x_1) \neq F$, $x_1 \in V_b$. Applying **Theorem 1**, there exists a Hamiltonian path $P(\phi(x_0), \phi(x_1))$ in $Q_{n-1}^1 - F^1 - \{a_0\}$. Thus, $\langle s \xrightarrow{P(s, x_0)} x_0, \phi(x_0) \xrightarrow{P(\phi(x_0), \phi(x_1))} \phi(x_1), x_1 \xrightarrow{P(x_1, y)} y, \phi(y) \rangle$ is a Hamiltonian path of $Q_n - F - \{a, b\}$ as illustrated in Figure 9.(e).

Suppose $\phi(b) = t$. Let $x_0, x_1 \in V_b$ and $\phi(x_0), \phi(x_1) \in V_w$ be fault-free vertices. Applying **Theorem 2**, we can obtain two spanning disjoint paths $P(s, x_0)$ and $\langle x_1 \xrightarrow{P(x_1, y)} y, b \rangle$ of $Q_{n-1}^0 - F^0$ for $\phi(y) \notin F^1$. Applying **Theorem 1**, we can construct a Hamiltonian path $P(\phi(x_0), \phi(x_1))$ of $Q_{n-1}^1 - F^1 - \{a, t, \phi(y)\}$. Thus, $\langle s \xrightarrow{P(s, x_0)} x_0, \phi(x_0) \xrightarrow{P(\phi(x_0), \phi(x_1))} \phi(x_1), x_1 \xrightarrow{P(x_1, y)} y, \phi(y), t \rangle$ is a Hamiltonian path of $Q_n - F - \{a, b\}$.

Case 2.2.3: $s \in Q_{n-1}^1$ and $t \in Q_{n-1}^0$.

Since $Q_{n-1}^0 - F^0$ is hyper-Hamiltonian laceable, there exist a Hamiltonian $P(t, x)$, and $\phi(x) \neq F$. Since $Q_{n-1}^1 - F^1$ is Hyper Hamiltonian laceable, there exist a Hamiltonian $P(\phi(x), s)$. Thus, $\langle s \xrightarrow{P(s, \phi(x))} \phi(x), x \xrightarrow{P(x, t)} t \rangle$ is a Hamiltonian path of $Q_n - F - \{a, b\}$, as illustrated in Figure 9.(f).

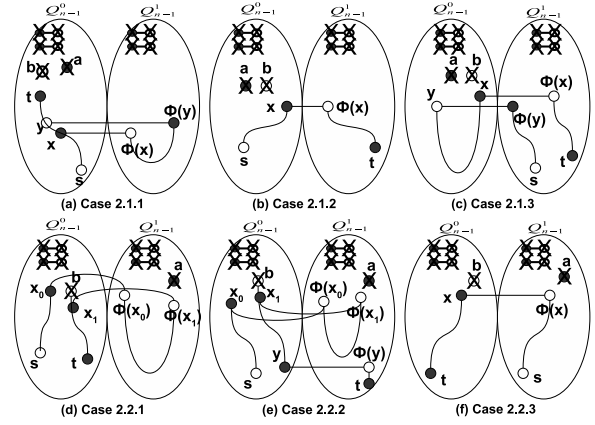


Figure 9: Illustration of Theorem 3.

5 The longest ring in faulty hypercube

In this section, we will construct the fault-free cycle in $Q_n - F_v$ where $4 \leq |F_v| \leq n - 1$, and the length of this cycle is at least $2^n - 2 |F_v| + 4$.

Theorem 4 *There exists a fault-free cycle of length at least $2^n - 2 |F_v| + 4$ in hypercube Q_n where F_v is the set of faulty vertices for $|F_v \cap V_b| \geq 2$, $|F_v \cap V_w| \geq 2$, $|F_v| \leq n - 1$ and $n \geq 5$.*

Proof: Let $x_0, x_1 \in F_v$ and $y_0, y_1 \in F_v$ be black and white vertices of Q_n , respectively. By symmetry of hypercube, we can arrange vertices x_0 and x_1 in different Q_{n-1} subgraphs. Without loss of generality, we can assume that $x_0 \in Q_{n-1}^0$ and $x_1 \in Q_{n-1}^1$. Let $F_v - \{x_0, x_1, y_0, y_1\} = \{a_1, a_2, \dots, a_{f-4}\}$ and $F_v' = \{b_i \mid (b_i, a_i) \text{ be an edge for } a_i \in F_v - \{x_0, x_1, y_0, y_1\}, \text{ and } b_i \text{ and } a_i \text{ are in the same subgraph}\}$. We will construct a Hamiltonian cycle of $Q_n - F_v$ for $n \geq 5$ in the following cases. The length of this cycle is $2^n - 2 |F_v| + 4$.

Case 1: $y_0, y_1 \in Q_{n-1}^0$.

Applying **Theorem 3**, we can obtain a Hamiltonian path $\langle w_0 \xrightarrow{P(w_0, w_1)} w_1, y_1 \rangle$ of $Q_{n-1}^0 - F_v - F_v'$ for $w_0, w_1 \in V_b$ and $\phi(w_0), \phi(w_1) \notin F_v \cup F_v'$. Applying **Theorem 1**, We can construct a Hamiltonian path $P(\phi(w_1), \phi(w_0))$ of $Q_{n-1}^1 - F_v - F_v'$. Thus, $\langle w_0 \xrightarrow{P(w_0, w_1)} w_1, \phi(w_1) \xrightarrow{P(\phi(w_1), \phi(w_0))} \phi(w_0), w_0 \rangle$ is a Hamiltonian cycle of $Q_n - F_v - F_v'$.

Case 2: y_0, y_1 in different Q_{n-1} subgraphs.

Without loss of generality, we assume that

$y_0 \in Q_{n-1}^0$ and $y_1 \in Q_{n-1}^1$. Let $w_0 \in V_b$, $w_1 \in V_w$, $\phi(w_0)$ and $\phi(w_1)$ be fault-free vertices. Applying **Theorem 3**, there exist a Hamiltonian path $P(w_0, w_1)$ and $P(\phi(w_1), \phi(w_0))$ of $Q_{n-1}^0 - F_v - F_v'$ and $Q_{n-1}^1 - F_v - F_v'$, respectively. Thus, $\langle w_0 \xrightarrow{P(w_0, w_1)} w_1, \phi(w_1) \xrightarrow{P(\phi(w_1), \phi(w_0))} \phi(w_0), w_0 \rangle$ is a Hamiltonian cycle of $Q_n - F_v - F_v'$.

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6 Conclusion

In this paper, we construct the fault-free cycle with at least $2^n - 2 |F_v| + 4$ in $Q_n - F_v$ where F_v is faulty vertices set with at least two black vertices and two white vertex for $4 \leq |F_v| \leq n - 1$. It is worth while to investigate the fault-free paths of faulty hypercube networks.

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